

Methods

Neural Efficiency Index

The neural efficiency (NE) index was computed following the framework proposed by Curtin and Ayaz (2019), adapted for the longitudinal fNIRS context. The index integrates behavioural performance (reading speed) and cortical activation (HbO beta coefficients) into a single composite score reflecting the efficiency of neural resource deployment during reading.

First-level GLM beta coefficients (HbO) were aggregated per participant, session, and ROI by averaging across all channels within each ROI (Frontal: 7 channels; Temporal: 9 channels). Beta coefficients were then standardised (z-scored) separately for each ROI, as frontal and temporal activations operate on different amplitude scales. Reading speed (TimeRm, in ms per character; lower values indicate faster reading) was standardised once over the full training sample (sessions 1-9, all 14 participants). The NE index was computed as:

$$NE = (-z_TimeRm - z_beta) / \sqrt{2}$$

The negative sign on z_TimeRm reflects the direction of the reading time variable: because lower TimeRm values indicate better (faster) performance, negating z_TimeRm ensures that both components increase with better performance and lower activation. The denominator $\sqrt{2}$ rescales the composite to unit variance. Higher NE values indicate a more favourable combination of faster reading with lower neural activation. A calibration-session baseline (`sessao_num = 0`) was used to compute per-participant `beta_baseline` and `TimeRm_baseline` values, which served as covariates in subsequent models. The NE index was computed for training sessions 1-9 only ($N = 252$ observations: 14 subjects x 9 sessions x 2 ROIs).

Linear Mixed Model — Neural Efficiency

Longitudinal changes in NE were modelled separately for each ROI using linear mixed models (LMMs) implemented in Python 3.12 with `statsmodels 0.14.6` (`smf.mixedlm` function). The model formula was:

$$NE \sim \text{group} * \text{sessao_num} + \text{TimeRm_baseline} + (1 \mid \text{subject})$$

where `group` (reference: `nao_acelerado`) and `sessao_num` (integer, 1-9) were fixed effects, their interaction term captured training-induced differential gains in NE, `TimeRm_baseline` was included as a covariate to control for pre-existing differences in reading speed, and a random intercept per subject accounted for between-subject variability. Models were estimated using restricted maximum likelihood (REML). Each ROI model contained 126 observations (14 subjects x 9 sessions).

Individual Neural Efficiency Regressions

To characterise individual-level NE trajectories, simple OLS regressions (NE ~ sessao_num) were estimated for each participant-ROI combination (N = 9 data points per regression). The regression slope quantifies the rate of NE change per session. Individual slopes were compared between groups using independent-samples t-tests and Fisher exact tests on the proportion of participants with significant slopes ($p < .05$).

Mann-Whitney Group Comparison of Mean NE

To assess overall group differences in NE magnitude, mean NE was computed per participant for each ROI by averaging across training sessions 1-9. Mann-Whitney U tests (two-tailed) compared the group distributions. Effect size was quantified as the rank-biserial correlation $r = 1 - (2U / n1*n2)$.

Results

Neural Efficiency LMM — Frontal ROI

For the Frontal ROI (N = 126), no fixed effects reached statistical significance. The group main effect was $\beta = 0.091$, $SE = 0.352$, $z = 0.258$, $p = .796$, 95% CI [-0.598, 0.780]. The session progression coefficient was $\beta = 0.046$, $SE = 0.035$, $z = 1.297$, $p = .194$, 95% CI [-0.023, 0.114]. The critical group x session interaction was $\beta = 0.067$, $SE = 0.050$, $z = 1.352$, $p = .176$, 95% CI [-0.030, 0.164], indicating no significant differential NE trajectory in the frontal cortex. The TimeRm_baseline covariate was statistically significant ($\beta = -0.012$, $SE = 0.003$, $z = -4.053$, $p < .001$, 95% CI [-0.018, -0.006]), indicating that faster baseline readers showed lower NE values during training.

Neural Efficiency LMM — Temporal ROI

For the Temporal ROI (N = 126), the group x session interaction was statistically significant: $\beta = 0.109$, $SE = 0.047$, $z = 2.298$, $p = .022$, 95% CI [0.016, 0.202]. This indicates that the Accelerated group showed a significantly steeper increase in temporal NE across training sessions relative to the Non-Accelerated group. The TimeRm_baseline covariate was also significant ($\beta = -0.010$, $SE = 0.003$, $z = -2.907$, $p = .004$, 95% CI [-0.017, -0.003]). No other terms reached significance.

Table 1. LMM Results for Neural Efficiency

Panel A - Frontal ROI (N = 126)

Parameter	Beta	SE	z	p	95% CI
Intercept	1.131	0.500	2.263	.024	[0.151, 2.110]
Group (Accelerated)	0.091	0.352	0.258	.796	[-0.598, 0.780]
Session	0.046	0.035	1.297	.194	[-0.023, 0.114]
Group x Session	0.067	0.050	1.352	.176	[-0.030, 0.164]
TimeRm_baseline **	-0.012	0.003	-4.053	<.001	[-0.018, -0.006]

Panel B - Temporal ROI (N = 126)

Parameter	Beta	SE	z	p	95% CI
Intercept	0.906	0.562	1.612	.107	[-0.195, 2.007]
Group (Accelerated)	-0.044	0.371	-0.119	.906	[-0.771, 0.683]
Session	0.027	0.034	0.810	.418	[-0.039, 0.093]
Group x Session *	0.109	0.047	2.298	.022	[0.016, 0.202]
TimeRm_baseline **	-0.010	0.003	-2.907	.004	[-0.017, -0.003]

* $p < .05$; ** $p < .001$. REML estimation. Reference group = Non-Accelerated.

Mann-Whitney Group Comparison of Mean NE

Comparison of mean NE across training sessions revealed significantly higher NE in the Accelerated group in both ROIs. Frontal ROI: Accelerated $M = 0.470$ ($SD = 0.286$) vs Non-Accelerated $M = -0.470$ ($SD = 0.762$); $U = 46$, $p = .004$, $r = -0.878$ (large). Temporal ROI: Accelerated $M = 0.543$ ($SD = 0.482$) vs Non-Accelerated $M = -0.543$ ($SD = 0.725$); $U = 47$, $p = .002$, $r = -0.918$ (large).

Individual NE Slopes

Individual OLS regressions were estimated for each participant x ROI combination. In the Frontal ROI, mean slope for the Accelerated group was 0.113 ($SD = 0.075$; 1/7 significant at $p < .05$) and for Non-Accelerated 0.046 ($SD = 0.107$; 1/7 significant). In the Temporal ROI, Accelerated mean slope was 0.136 ($SD = 0.109$; 3/7 significant) and Non-Accelerated 0.027 ($SD = 0.109$; 2/7 significant). Between-group comparisons did not reach significance in either ROI (Frontal: $t p = .198$, Fisher $p = 1.0$; Temporal: $t p = .086$, Fisher $p = 1.0$). Individual slopes are presented in Table 2.

Table 2. Individual Neural Efficiency Slopes (NE ~ session, OLS)

Subject	Group	ROI	Slope	p-value	R2	Sig.
SUBJ 002	Accelerated	Frontal	0.193	0.016	0.590	$p < .05$
SUBJ 002	Accelerated	Temporal	0.106	0.085	0.364	$p < .10$
SUBJ 004	Accelerated	Frontal	0.196	0.229	0.199	n.s.
SUBJ 004	Accelerated	Temporal	0.257	0.001	0.811	$p < .05$
SUBJ 006	Accelerated	Frontal	-0.012	0.873	0.004	n.s.
SUBJ 006	Accelerated	Temporal	-0.003	0.979	0.000	n.s.
SUBJ 007	Accelerated	Frontal	0.107	0.656	0.030	n.s.
SUBJ 007	Accelerated	Temporal	0.232	0.034	0.497	$p < .05$
SUBJ 027	Accelerated	Frontal	0.053	0.389	0.108	n.s.
SUBJ 027	Accelerated	Temporal	0.198	0.002	0.755	$p < .05$
SUBJ 028	Accelerated	Frontal	0.110	0.061	0.415	$p < .10$
SUBJ 028	Accelerated	Temporal	0.173	0.108	0.327	n.s.
SUBJ 034	Accelerated	Frontal	0.142	0.053	0.437	$p < .10$
SUBJ 034	Accelerated	Temporal	-0.010	0.949	0.001	n.s.
SUBJ_001	Non-Accelerated	Frontal	0.068	0.354	0.123	n.s.

SUBJ_001	Non-Accelerated	Temporal	0.005	0.920	0.002	n.s.
SUBJ_005	Non-Accelerated	Frontal	0.003	0.945	0.001	n.s.
SUBJ_005	Non-Accelerated	Temporal	0.025	0.738	0.017	n.s.
SUBJ_018	Non-Accelerated	Frontal	0.078	0.138	0.286	n.s.
SUBJ_018	Non-Accelerated	Temporal	0.001	0.993	0.000	n.s.
SUBJ_024	Non-Accelerated	Frontal	0.115	0.204	0.219	n.s.
SUBJ_024	Non-Accelerated	Temporal	0.169	0.001	0.801	p < .05
SUBJ_030	Non-Accelerated	Frontal	-0.116	0.124	0.303	n.s.
SUBJ_030	Non-Accelerated	Temporal	-0.130	0.219	0.207	n.s.
SUBJ_035	Non-Accelerated	Frontal	-0.040	0.515	0.063	n.s.
SUBJ_035	Non-Accelerated	Temporal	-0.049	0.664	0.029	n.s.
SUBJ_037	Non-Accelerated	Frontal	0.210	0.017	0.582	p < .05
SUBJ_037	Non-Accelerated	Temporal	0.169	0.023	0.546	p < .05